

EGE UMUT CAN

Independent Student Researcher in Macroeconomics, Financial Econometrics & Market Microstructure

Istanbul, Türkiye | FMV Isik Schools (Class of 2027) | **Live Data Lab:** ege211.github.io/macroeconomic-data-lab | **GitHub:** github.com/ege211

EDUCATION

FMV Isik Schools, Istanbul, Türkiye

2023 – June 2027 (Expected)

- **Academic Standing:** Cumulative GPA: **90.0 / 100.0**
- **Advanced Placement (AP) Coursework:** AP Macroeconomics, AP Microeconomics, AP Calculus, AP Psychology
- **Research Focus:** Monetary Policy Credibility & Inflation Persistence, Capital Market Microstructure, Financial Machine Learning, Strategic Capital Allocation

FOUNDER & OPEN-SOURCE RESEARCH PLATFORM

Founder, Developer & Lead Student Researcher — Comparative Macroeconomic Data Lab (GEA)

2025 – Present

Interactive Web Laboratory: ege211.github.io/macroeconomic-data-lab | Open-Source Codebase: github.com/ege211/macroeconomic-data-lab

- **Live API Architecture:** Engineered an open-source macroeconomic web laboratory retrieving live **World Bank Open Data API v2** indicators (*NY.GDP.MKTP.KD.ZG*, *FP.CPI.TOTL.ZG*) across five economies (Türkiye, United States, Germany, United Kingdom, Japan; 2019–2023) with automated fallback verification and CSV data-cube export.
- **Dual Quantitative Simulators:** Built two interactive browser-based simulation engines in JavaScript and Chart.js: (1) a *2024–2025 Cross-Country Scenario Simulator* modeling heterogeneous GDP and CPI responses to hypothetical policy-rate (+0 to +500 bps) and supply-price (+0% to +20%) shocks; and (2) an *AR(1) Inflation Persistence & Shock Half-Life Simulator* operationalizing Working Paper WP-01 (beta = 0.72, shock half-life $t_{1/2} = 2.11$ years vs. 0.58 years under anchored expectations).
- **10-Paper Research Corpus & Intellectual Map:** Authored and curated 10 full-text working papers, empirical essays, and methodological research notes organized across four thematic pillars, accompanied by an annotated theoretical reading list linking Acemoglu & Robinson, Gali & Gertler (1999), Taylor (1993), Piketty (2014), SPK Communique III-52.1, and Porter (1996).

INDEPENDENT ECONOMETRIC & QUANTITATIVE WORKING PAPERS

A Machine-Learning-Based Early Warning Framework for Systemic Banking Crises

2026

- **Panel Architecture:** Evaluated macro-financial vulnerability indicators across a balanced panel of **76 World Bank economies (1990–2025; 684 out-of-sample evaluations)** using penalized Logistic Regression and Random Forest classifiers.
- **Leak-Free Temporal Validation:** Constructed an expanding-window validation pipeline ($T = 2000–2008$), isolating train-only preprocessing and cost-sensitive class weighting to address a **2.78% rare-event crisis base rate**.
- **Empirical Diagnostic:** Demonstrated that adding balance-sheet leverage (private credit-to-GDP) and external deficits more than doubled pooled crisis recall to **47.37%** (vs. 21.05% baseline), while diagnosing an out-of-sample cross-sectional ranking breakdown (**ROC-AUC = 0.2301**) outside the 2007 pre-GFC synchronization cluster.

AI-Assisted Equity Valuation: A Walk-Forward Study on SEC Form 10-K Filings

2026

- **Point-in-Time Data Engineering:** Extracted 12 pre-specified linguistic sentiment and complexity dimensions across **30 non-financial S&P large-cap firms (2014–2024)** via an automated five-tier SEC EDGAR XBRL normalization cascade.
- **Structural DCF & Audit Bridge:** Built a deterministic Discounted Cash Flow (DCF) and Reverse DCF valuation engine linked to an auditable qualitative translation bridge (*valuation_bridge_v1.0*) enforcing bounded parameter adjustments (e.g., -120 bps operating margin haircut).
- **Null-Result Proof:** Proved via walk-forward out-of-sample testing that 10-K filing text signals yield **zero statistically significant incremental predictive power** over trailing accounting fundamentals in forward margin forecasting ($p = 0.2335$), with unconstrained high-dimensional text features inducing a **-13.14% performance degradation**.

Strategic Capital Allocation in Investment Banking: G-SIBs vs. Advisory Boutiques

2026

- **Primary Archival Dataset:** Hand-collected **540 audited firm-year observations across 40 SEC Form 10-K filings (2020–2024)** to compare capital deployment across Global Systemically Important Bank Holding Companies (Panel A) and Independent Advisory Boutiques (Panel B).
- **Cross-Sectional Findings:** Disclosed **\$233.2B in shareholder distributions** (\$113.8B dividends, \$119.3B repurchases) and identified diverging lagged dynamics: near-zero correlation with subsequent ROE in BHCs ($r = -0.049$) vs. negative correlation with operating margins in boutiques ($r = -0.403$).
- **Institutional Framework:** Formulated an author-led conceptual model showing how Basel III regulatory constraints (**CET1 buffers of 13.1%–17.4%**) vs. talent-retention liquidity needs dictate diverging capital-allocation functions.

Inflation Persistence and Monetary Policy Effectiveness: Cross-Country Panel Evidence (WP-01)

2026

- **Dynamic Panel Specification:** Estimated autoregressive panel regressions across **45 advanced and emerging economies (2000–2023)** using World Bank, IMF IFS, and BIS policy-rate data.
- **Key Findings:** Documented high inflation persistence ($\beta = 0.72$) in emerging markets and a 2–4 quarter monetary policy transmission lag ($\gamma < 0$), showing that central bank credibility anchors expectations and compresses shock half-lives.

Sports Finance & Tactical Analytics: Arsenal FC Longitudinal Study (2015/16–2024/25)

2026

- **Multi-Source Integration:** Integrated audited corporate accounts (*Arsenal Holdings Limited*), a **128-transaction transfer census**, and advanced tactical metrics (*Understat xG / xPTS*) across 10 completed seasons.
- **Process vs. Outcome Variance:** Established that transfer investments correlate significantly with underlying tactical process metrics ($xGD\ r = +0.5982$; $xPTS\ r = +0.6189$), whereas realized match points ($PPG\ r = +0.3911$) reflect higher stochastic variance.
- **Wage-Bill Counter-Evidence:** Challenged conventional wage-dominance theories in single-club settings by revealing near-zero, sign-reversing associations between statutory staff costs and subsequent points ($r = +0.0516$; Spearman $\rho = -0.0333$).

MACROECONOMIC POLICY, MARKET MICROSTRUCTURE & STRATEGY PAPERS

- **Borsa Istanbul — The Fragility of Liquidity (2026):** Investigated capital market microstructure in Türkiye across the 2022–2023 retail equity inflow and May 2026 liquidity contraction; analyzed mutual fund redemption cascades under **SPK Communiqué III-52.1** and asymmetric risk pricing in corporate debt (*tahvil*).
- **Exchange Rate Pass-Through (ERPT) in Emerging Markets (2026):** Contrasted short-run import-price pass-through elasticities in emerging markets (**0.38**) vs. advanced economies (**0.09**), examining non-linear threshold effects during rapid currency depreciations.
- **The Credibility Gap & The Lira Doesn't Lie (2026):** Examined central bank independence, time-inconsistency, behavioral anchoring breakdown, and household dollarization in Türkiye as an endogenous response to institutional volatility.
- **The Paradox of Petrowealth & Contested Terrain (2026):** Comparative political-economy essays analyzing Dutch disease and sovereign wealth governance in Venezuela vs. Norway/GCC, alongside foundational debates in minimum-wage elasticity (Card & Krueger), stakeholder governance, and Piketty's $r > g$.
- **Strategic Case Study — Inditex (Zara) Operating Architecture (2026):** Analyzed Inditex's integrated operating system via Porter's Five Forces, VRIO, and Value Chain frameworks, evaluating how 2–3 week design-to-retail lead times mitigate inventory risk while facing C2M disruption (Shein/Temu) and sustainability trade-offs.
- **Data Lab Research Notes I & II (2026):** Authored two companion methodological notes—"One Shock, Five Trajectories (2019–2023)" and "Assumptions Made Visible"—explicitly documenting parameter uncertainty, the Lucas Critique, and identification limits in calibrated macroeconomic simulators.

FIELD OBSERVATIONS & INSTITUTIONAL SIMULATIONS

Operations & Market Strategy Field Researcher — Bohoo Textile & Corporate Fleets

Istanbul & London

- Analyzed B2B international textile export workflows, customs documentation, and customer value propositions (CVP), documenting how relationship trust and lead-time guarantees insulate mid-market suppliers from pure price competition.
- Executed observational operational studies across contrasting business models: administrative capacity bottlenecks in B2B corporate fleet leasing vs. real-time service-profit chains in UK hospitality.

Institutional Financial Markets Job Simulations

Forage / Virtual Programs

- **Bank of America Global Markets (Sales & Trading):** Evaluated macroeconomic market catalysts, institutional client positioning, and trading-desk process automation.
- **Bank of America Private Bank:** Formulated multi-generational client wealth allocation and strategic alignment proposals.
- **Goldman Sachs Risk Analysis:** Conducted quantitative credit and commercial real-estate asset risk profiling under downside stress scenarios.

TECHNICAL SKILLS & LANGUAGES

- **Quantitative & Programming:** Python (*pandas, scikit-learn, numpy, statsmodels, pytest*), SQL / DuckDB, Panel & Time-Series Econometrics, Walk-Forward Backtesting, Out-of-Sample Validation.
- **Financial & Web Engineering:** SEC EDGAR XBRL Parsing, Deterministic & Reverse DCF Modeling, World Bank REST API v2 Integration, Git / GitHub, HTML5 / CSS3 / JavaScript (*Chart.js*).
- **Languages:** Turkish (Native), English (C1 / Advanced Academic Proficiency).